1.23 Graph Sketching: Hyperbolas

Worked Solutions

1 a)

$\frac{\sqrt{4}}{\sqrt{16}} = \frac{2}{4} = \frac{1}{2}$

$$y = \frac{1}{2}x \text{ and } y = -\frac{1}{2}x$$

1 b)

$x^2 - 9y^2 = 36 \Rightarrow \frac{x^2}{36} - \frac{y^2}{4} = 1$

$$\frac{\sqrt{4}}{\sqrt{36}} = \frac{2}{9} = \frac{1}{3}$$

$y = \frac{1}{3}x$ and $y = -\frac{1}{3}x$

1 c)

$x^2 - 2y^2 = 16 \Rightarrow \frac{x^2}{16} - \frac{y^2}{8} = 1$

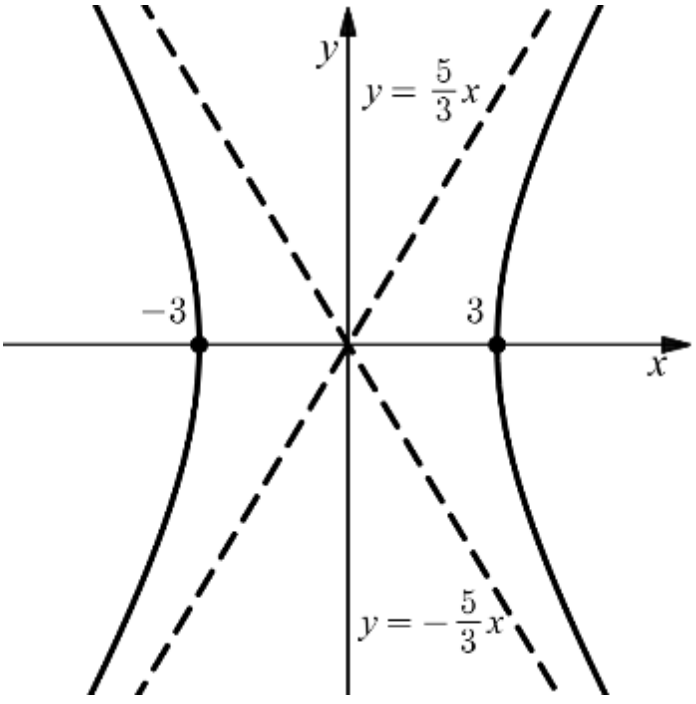
$$\frac{\sqrt{8}}{\sqrt{16}} = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2}$$

$y = \frac{\sqrt{2}}{2}x$ and $y = -\frac{\sqrt{2}}{2}x$

2 a)

$\frac{\sqrt{25}}{\sqrt{9}} = \frac{5}{3}$ so asymptotes $y = \pm\frac{5}{3}x$

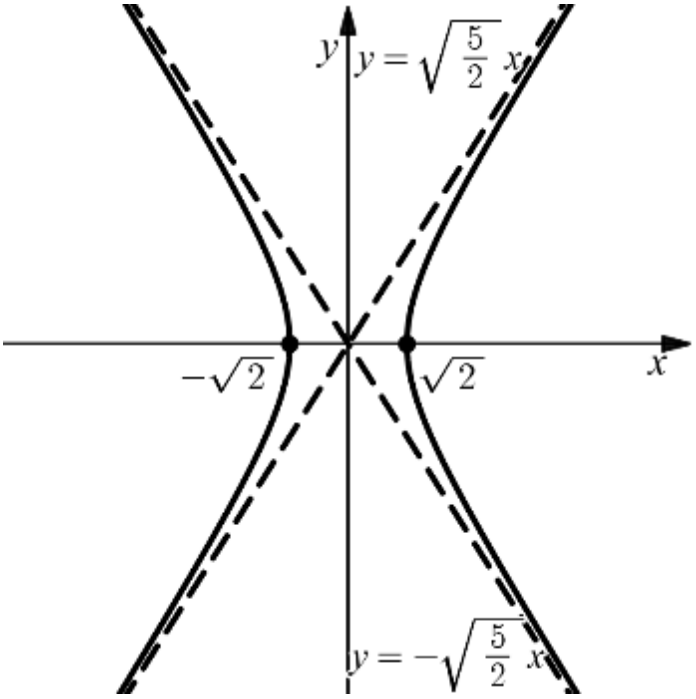
When $y = 0$, $x = \pm 3$



2 b)

$\frac{\sqrt{5}}{\sqrt{2}} = \sqrt{\frac{5}{2}}$ so asymptotes $y = \pm\sqrt{\frac{5}{2}}x$

When $y = 0$, $x = \pm\sqrt{2}$

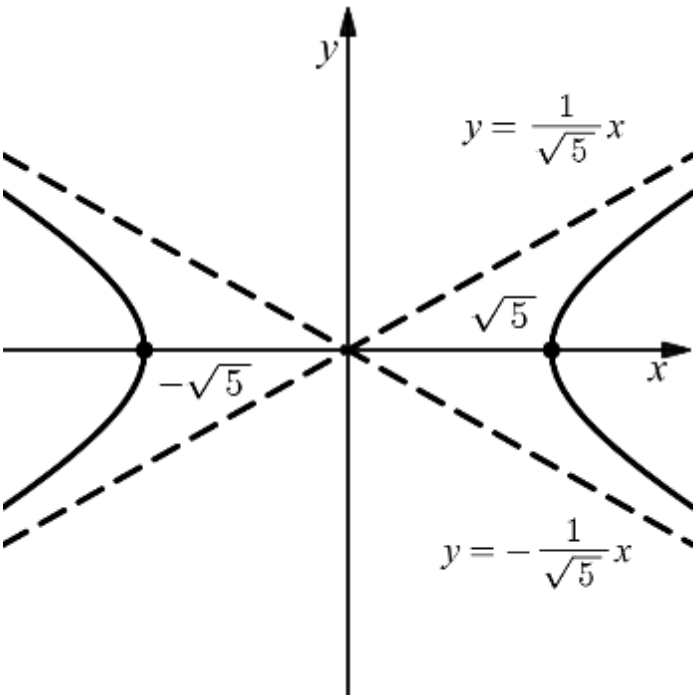


2 c)

$x^2 - 5y^2 = 5 \Rightarrow \frac{x^2}{5} - y^2 = 1$

$\frac{1}{\sqrt{5}}$ so asymptotes $y = \pm\frac{1}{\sqrt{5}}x$

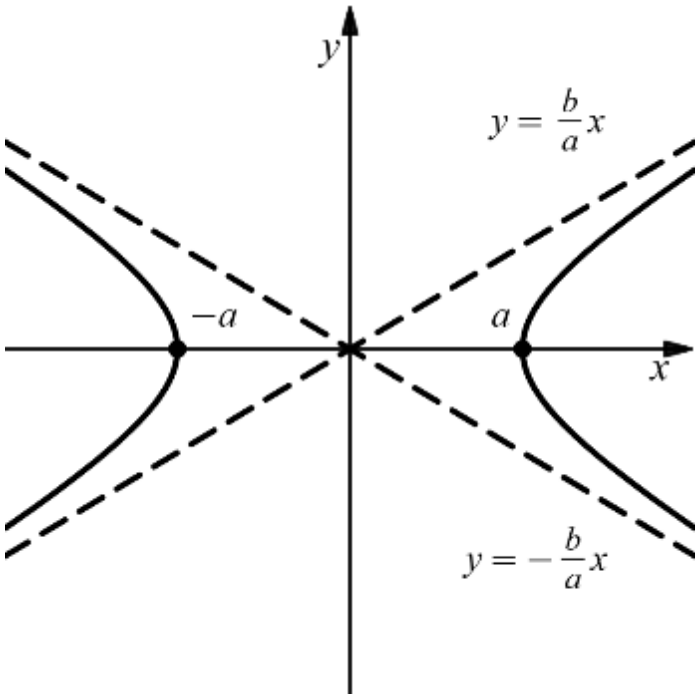
When $y = 0$, $x = \pm\sqrt{5}$



2 d)

$\frac{b}{a}$ so asymptotes $y = \pm\frac{b}{a}x$

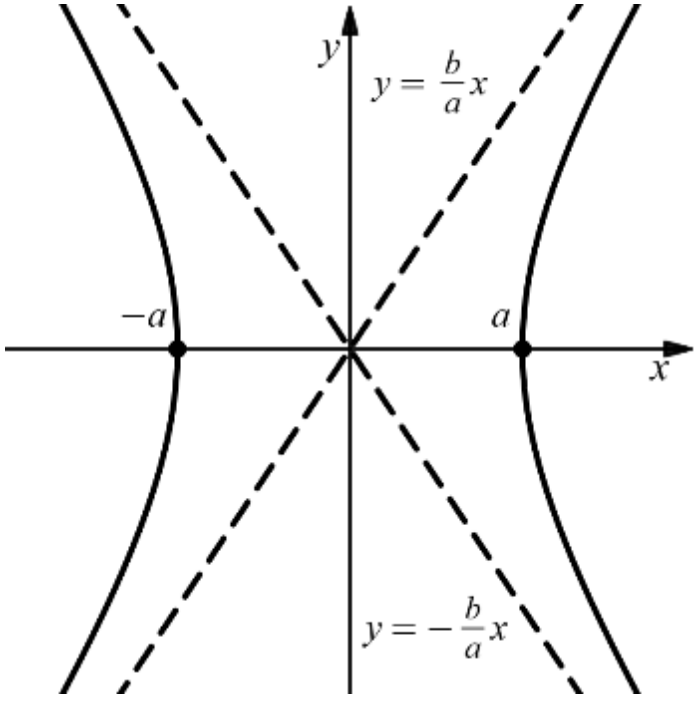
When $y = 0$, $x = \pm a$



2 e)

$\frac{b}{a}$ so asymptotes $y = \pm\frac{b}{a}x$

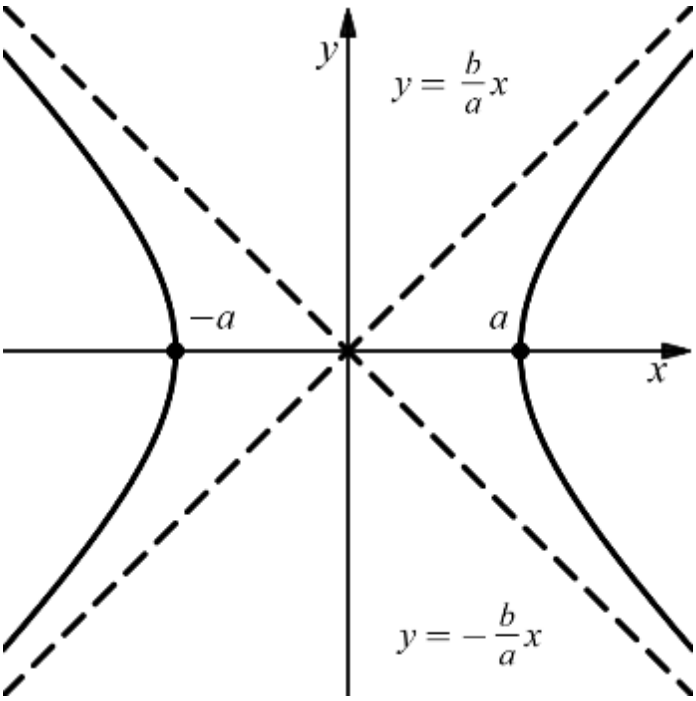
When $y = 0$, $x = \pm a$



2 f)

$\frac{b}{a}$ so asymptotes $y = \pm\frac{b}{a}x$

When $y = 0$, $x = \pm a$



3 a)

$\frac{b}{a} = \frac{2}{3}$ and $a = 3$

$\therefore a = 3$ and $b = 2$

3 b)

$\frac{b}{a} = 2$ and $a = 2$

$\therefore a = 2$ and $b = 4$

3 c)

$\frac{b}{a} = \sqrt{3}$ and $a = \sqrt{7}$

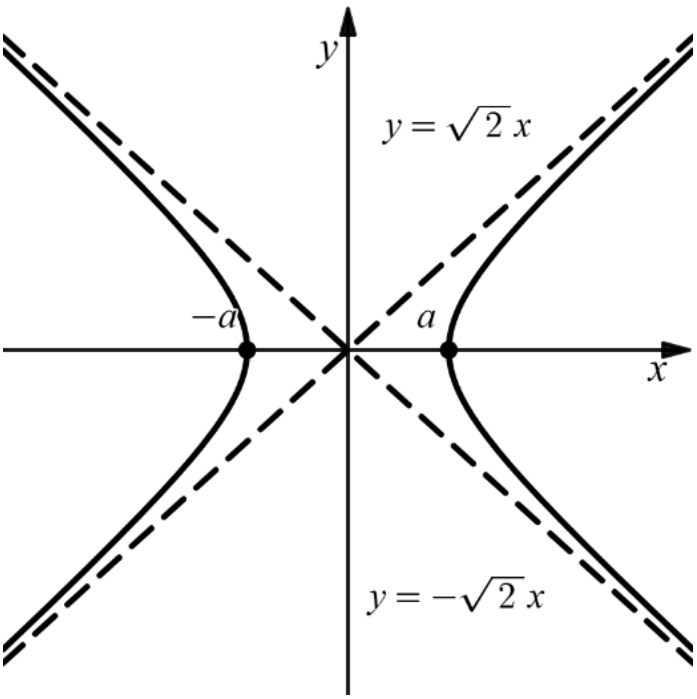
$\therefore a = \sqrt{7}$ and $b = \sqrt{21}$

4 a)

$y = \pm\frac{2a^2}{a^2}x$

Asymptotes are $y = \sqrt{2}x$ and $y = -\sqrt{2}x$

4 b)



5 a)

At $(-3, 0)$

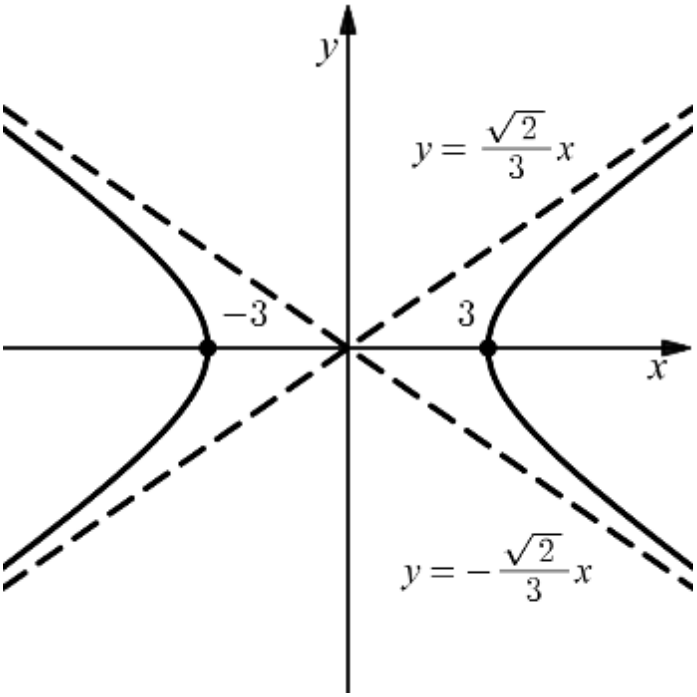
$$\frac{9}{a^2} = 1 \text{ so } a = \pm 3$$

At $(6, \sqrt{6})$

$$\frac{36}{9} - \frac{6}{b^2} = 1 \text{ so } b = \pm\sqrt{2}$$

5 b)

$\frac{x^2}{9} - \frac{y^2}{2} = 1$



6 a)

At $(2, -\sqrt{3})$

$$\frac{4}{a^2} - \frac{3}{b^2} = 1$$

At $(-3, 3\sqrt{2})$

$$\frac{9}{a^2} - \frac{18}{b^2} = 1$$

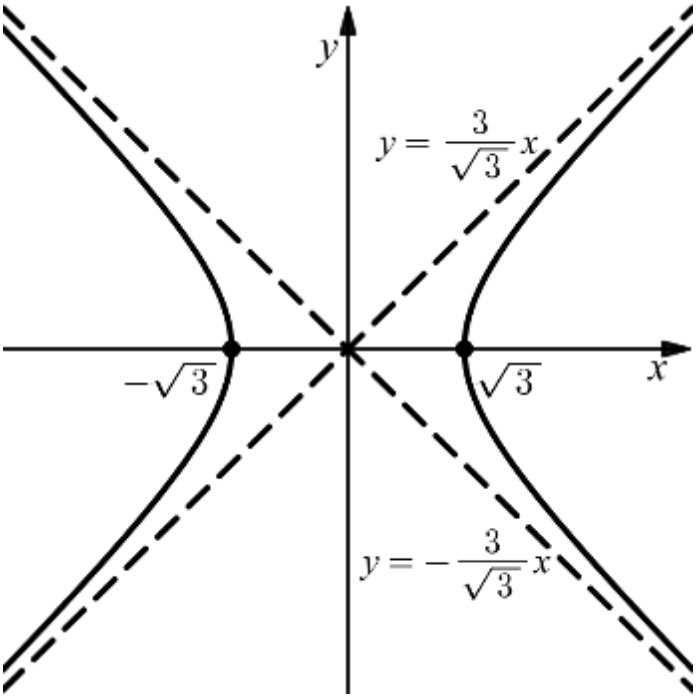
So that

$$\frac{24}{a^2} - \frac{18}{b^2} = 6 \Rightarrow \frac{24}{a^2} + (1 - \frac{9}{a^2}) = 6$$
$$\Rightarrow \frac{15}{a^2} = 5 \Rightarrow a^2 = 3 \quad \therefore a = \pm\sqrt{3}$$
$$\frac{4}{3} - \frac{3}{b^2} = 1 \Rightarrow b^2 = 9$$

$\therefore a = \pm\sqrt{3}$ and $b = \pm 3$

6 b)

$\frac{x^2}{3} - \frac{y^2}{9} = 1$



7)

Assume curve of the form:

$$\frac{y^2}{b} - \frac{x^2}{a} = 1$$

Let the asymptote be:

$$\frac{a}{b} = \frac{1}{2}$$

From the intercepts: $b = 2$ so $a = 4$

$$\frac{y^2}{4} - \frac{x^2}{16} = 1$$
$$-4y^2 + x^2 = -16$$
$$\therefore x^2 - 4y^2 = -16$$